Journal of Pain 40 (2026) 105644

Contents lists available at ScienceDirect

The Journal of Pain

journal homepage: www.jpain.org

Original reports

Are gray matter changes following pain reprocessing therapy associated with clinical improvement? A voxel-based morphometry mediation analysis

Carlos Murillo [a], Tor D. Wager [b], Yoni K. Ashar [a,*]

[a] Division of General Internal Medicine, School of Medicine, University of Colorado Anschutz Medical Campus, Aurora, United States
[b] Department of Psychological and Brain Sciences, Dartmouth College, Hanover, United States

ARTICLE INFO

Keywords:
Gray matter
Voxel-based morphometry
Mediation analysis
Pain Reprocessing Therapy
Chronic back pain

ABSTRACT

Chronic back pain (CBP) has been associated with reorganization of gray matter (GM), and treatment may reverse these GM adaptations. Prior trials testing treatment effects on GM did not included a randomized control group, found no treatment vs. control differences, or did not attempt to link treatment-induced GM changes to improvements in clinical outcomes. We addressed this gap by performing a voxel-based morphometry mediation analysis testing for effects of a psychological treatment (pain reprocessing therapy, PRT, n=44) vs. two randomized control groups (treatment as usual, n=47, and open-label placebo, n=44) on GM, and testing whether treatment-specific GM changes were associated with reductions in disability (ODI) and pain intensity (BPI). Relative to both control groups, PRT increased GM in the anterior cingulate and orbitofrontal cortex. These treatment-induced GM increases, however, did not mediate PRT effects on clinical outcomes. Instead, improvements in disability and pain were associated with GM increases in the lateral prefrontal cortex and GM decreases in the thalamus and amygdala. These findings indicate that PRT vs. control increases GM in key pain-modulatory regions, and, that clinical improvement is related to different prefrontal and subcortical GM changes in regions processing nociceptive input and regulating pain. The lack of mediation may indicate either that GM changes are not a treatment mechanism, or a lack of power to detect smaller effects. Future research is needed to better understand how treatments act on neurobiology to drive clinical improvement.

1. Introduction

Back pain is a highly prevalent disorder and a leading cause of disability worldwide.[1] Recurrence is common, and up to a third of the people develop chronic pain.[2,3] Chronic back pain (CBP) is a multifactorial problem characterized by a range of biophysical, psychological, and social dimensions that impair daily life.[4] People with CBP exhibit a structural reorganization of the brain, including gray matter (GM) changes in brain regions involved in top-down modulation of pain (e.g., anterior cingulate, anterior insula, and prefrontal cortex) and GM alterations in subcortical structures receiving nociceptive input (e.g., thalamus, amygdala).[5–8] These GM adaptations have been associated with patients' level of pain and disability,[9–13] and when symptoms remit, either spontaneously or after treatment, GM changes may be at least partially reversed.[14–17]

In recent years, randomized controlled trials have examined whether GM changes follow psychological interventions. These therapies target fear, avoidance, and negative pain beliefs that perpetuate disability and pain.[18–21] These maladaptive responses amplify pain by influencing its top-down regulation and are associated with GM reductions in key modulatory and attentional regions.[10,11,22–25] Thus, psychological therapies may reverse the GM adaptations observed in people with chronic pain. However, trials to date examining the effects of pain treatments on GMV have faced several limitations, including small sample sizes[16,26–34] or a lack of control groups.[35–37] Randomized trials to date including active control groups have largely reported no treatment-specific effects on GMV.[25,38,39] Studies are needed that have larger sample sizes and randomized controlled groups, with analytic plans that link treatment-induced changes in GMV to improvements in clinical outcomes. Linking treatment-induced GMV changes to clinical improvement is critical for identifying treatment mechanisms—how treatment changes the brain to relieve chronic pain.

Mediation analysis provides a method to jointly examine whether a treatment changes target mechanisms and whether these changes result in improved treatment outcomes.[40] To date, mediation analysis in pain treatment has been largely limited to tests of self-reported measures or

* Correspondence to: University of Colorado Anshutz Medical Campus, 13001 E 17th Pl, Aurora, CO 80045, United States.
E-mail address: yoni.ashar@cuanschutz.edu (Y.K. Ashar).

https://doi.org/10.1016/j.jpain.2025.105644
Received 14 May 2025; Received in revised form 19 November 2025; Accepted 11 December 2025
Available online 23 December 2025
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clinical features as mediators.[41–43] Mediation analysis of neuroimaging features has recently emerged in pain research to help identify brain mechanisms of acute pain modulation.[44–48] To our knowledge, mediation has not been used to test whether treatment relative to control affects GMV to improve chronic pain.

Pain reprocessing therapy (PRT) is a novel psychological therapy integrating pain neuroscience education, interoceptive exposures, and other techniques to help patients recover from chronic pain. PRT aims to reverse the nociplastic changes driving primary pain conditions like non-specific CBP. In an initial trial, PRT lead to large reductions in pain intensity, reduced functional brain responses to evoked back pain in the insula and midcingulate, and increase resting connectivity from the insula and prefrontal cortex to the somatosensory cortex and between midcingulate and precuneus.[49,50] These reductions in pain and functional brain changes raise the question of whether PRT might also reverse the structural brain changes observed in cross-sectional studies of patients vs. healthy controls. No prior studies have examined PRT effects on GM. The present study tested the hypothesis that one mechanism by which PRT improves pain is by inducing GMV changes.

We examined (i) whether PRT impacted GM compared to treatment as usual (TAU) and a placebo control, (ii) how GM changes related to improvements in clinical outcomes across groups, and (iii) the spatial overlap (across brain regions) between these two effects (mediation effect). As prior meta-analyses identified CBP vs. control GM differences largely in the prefrontal cortex, ventral striatum, amygdala, and thalamus,[5–7] we hypothesized that GM changes would be identified in these regions, and that identified treatment-induced GM changes would mediate PRT effects on clinical outcomes.

2. Methods

2.1. Study design and participants

This was a randomized trial (NCT03294148) conducted from August 2017 to November 2018 in Boulder, Colorado, USA. PRT was compared to open-label placebo and TAU in people with CBP.[49] Hundred fifty-one participants with low back pain for at least half the days during the last 6 months and 1-week average pain intensity score of $\geq$ 4/10 were enrolled. Individuals were excluded based on self-reported diagnoses or conditions including inflammatory disorders, metastasizing cancer, significant unexplained weight loss, bowel or bladder dysfunction, severe psychiatric disorders, intravenous drug use, logistical or technical barriers to participation, recent pain-related compensation or litigation, MRI contraindications, and neurological histories (e.g., stroke, brain surgery, brain tumor). Participants were recruited from the community using electronic advertisements and posters, social media, and referrals. Participants were randomized to each arm with equal probability, balancing on age, sex, pain intensity, and opioid use (yes/no) using an imbalance minimization algorithm.[51] Researchers involved in data collection were blind to group assignment. MRI data and clinical outcomes were collected at pre-treatment before randomization (i.e., T0) and immediately post-treatment (i.e., T1, approximately 5 weeks after randomization). Clinical outcomes were collected at the same timepoints. As this mediation analysis was not planned *a priori*, sample size of this secondary analysis was not informed by a power calculation.

Results from the test for treatment effects on resting connectivity and brain responses to evoked back pain are reported somewhere else.[49] Treatment effects on GM have not been previously published. This study was approved by the Ethical Committee at University of Colorado and all procedures were performed in accordance with the Declaration of Helsinki. All participants provided written informed consent prior to participation. Research methods and reporting were in accordance with current voxel-based morphometry (VBM) guidelines[52] and the AGReMA[53] guidelines for mediation analysis.

2.2. Interventions

Participants randomized to PRT completed 8 1-h individual therapy sessions twice weekly for 4 weeks. PRT sessions focused on (i) offering personalized evidence to explain centralized pain; (ii) guiding patients to reassess their pain sensations as non-dangerous both while seated and during feared postures or movements; (iii) addressing psychosocial factors, such as difficult emotions, exacerbating pain; and (iv) promoting strategies to enhance positive emotions and self-compassion. Further information on the PRT protocol can be found in the supplementary materials of the published primary trial.[49]

Participants assigned to the placebo condition received a saline subcutaneous injection at the most painful back site during an empathic, validating clinical visit at an orthopedic medical center. Prior to the administration of the placebo, participants watched two videos describing how placebo treatments can powerfully relieve pain when known to be inert. This condition was designed to provide at least partial control for patient expectations, an empathic encounter with a provider, and some elements of pain neuroscience education. Open-label placebo treatments have shown to be as effective or nearly as effective as traditional (deceptive) placebos.[54,55]

Participants assigned to TAU were given no additional treatment. Participants in TAU and both other conditions agreed to maintain their current care routine without initiating any new treatments (other than study-related procedures) until after the study period.

The PRT providers were experts in PRT prior to study initiation, and no provider training was required. Fidelity assessment, conducted via video-review of therapy sessions, found high fidelity delivery of PRT (see [49] for details).

2.3. Assessments

For mediation analyses, pre-to-post-treatment changes in GMV was the mediator under investigation (see below). Disability and average pain intensity measured at post-treatment were the outcomes of interest. Pain-related disability was assessed with the Oswestry Disability Index (ODI), which has shown good internal consistency ($\alpha = 0.76 – 0.86$) and test–retest reliability ($r = 0.83 – 0.91$).[56] The ODI ranges from 0 to 100, where higher scores indicate more disability. Average pain intensity was measured with numerical rating scale from 0 to 10 from the Brief Pain Inventory (BPI) Short Form. This scale has shown good internal consistency ($\alpha = 0.82 – 0.89$) and test–retest reliability ($r = 0.67 – 0.82$).[57]

2.3.1. MRI data acquisition

MRI data were collected using a 3 T MRI scanner (Siemens MAGNETOM Prisma) using a 32-channel head coil. Structural images were acquired using a single shot T1 MPRAGE sequence with repetition time = 2.4 s, echo time = 2.07 ms, flip angle = 8°, number of slices = 224, slice orientation = sagittal, voxel size = 0.8 mm isotropic, field of view = 256 × 256 mm^2, GRAPPA acceleration factor = 2.

2.3.2. Voxel-based morphometry

Voxel-based morphometry (VBM)[58] was implemented using the computational anatomy toolbox (CAT12)[59] standard pre-processing pipelines for longitudinal analysis, optimized to detect small changes over time. Each participant's T1-weighted image from pre-treatment and post-treatment was realigned using inverse-consistent rigid-body registrations and corrected for intra-subject bias-field inhomogeneities.[60,61] The generated images were then corrected for bias-field inhomogeneities and segmented into GM, white matter, and cerebrospinal fluid (see Gaser, Dahnke, Thompson, Kurth and Luders [59] for more details on CAT12's tissue segmentation step). Individual GM segments were spatially normalized into standard MNI space using Geodesic Shooting[62] and modulated by the Jacobian determinants from the corresponding flow fields to restore the volumetric information lost during the high-dimensional spatial registration. This modulation step

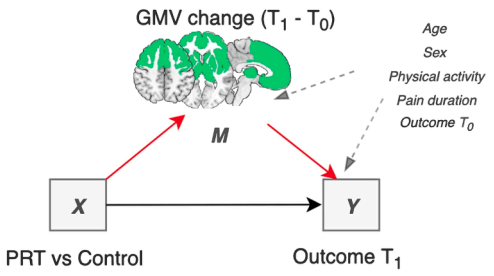


Fig. 1. Hypothesized directed acyclic graph depicting the mediation model. Red arrows depict the indirect effect from PRT (X), compared to either control therapy (TAU or Placebo), to the outcome (Y) through GMV changes in the prefrontal cortex, the insula and several subcortical structures (mediator; M). The black arrow depicts the total effect and the remaining (direct) effect from PRT vs control to the outcome. Grey dashed arrows depict confounding of the mediator-outcome relationship by pre-treatment variables. The hypothesized pre-treatment confounders were age, sex, physical activity, pain duration, as well as the baseline value of the outcome. PRT, Pain Reprocessing Therapy; GMV, Gray Matter Volume; T1, post-treatment; T0, pre-treatment.

involves multiplying each voxel by the relative change in volume, which allows for a comparison of absolute GM volume (GMV) corrected for individual brain size.[63] Finally, modulated images were spatially smoothed (8 mm full width at half maximum Gaussian kernel). Though the choice of processing pipeline has a large impact on cross-sectional GMV analyses, the longitudinal test-retest reliability of validated pipelines like CAT is generally high.[64] This supports our analyses of pre-to-post-treatment changes in GMV from a single pipeline.

Quality inspection of the MRI images was performed in stages. Raw images were visually inspected for artifacts before pre-processing. Image quality ratings (IQR; a summary measure of noise, bias, and image resolution)[65] for each image were calculated after pre-processing and evaluated following CAT workflow.[59] Overall, the IQR scores were in the "good" range after pre-processing (81.43 ± (SD)2.04% at pre-treatment and 81.33 ± 2.40% at post-treatment, on a 100-point image quality scale), with no between-group differences in image quality.

2.4. Statistical analyses

A complete-cases approach was adopted for the mediation analysis. The confounding assumptions for mediation analysis are critical, as their violations can bias the results.[40] Figure 1 shows a causal directed acyclic graph representing the causal assumptions of the mediation analysis. Sex, age, physical activity and pain duration (in years) were included to control for treatment-mediator and mediator-outcome confounding, given their relationship with GM and the outcome.[12,66–70] The pre-treatment value of the outcome was also included as a potential confounder.[71] Analyses were conducted using the Canlab Mediation Toolbox[72] (https://github.com/canlab/MediationToolbox), implemented in MATLAB (version R2023b, MathWorks). To limit the search

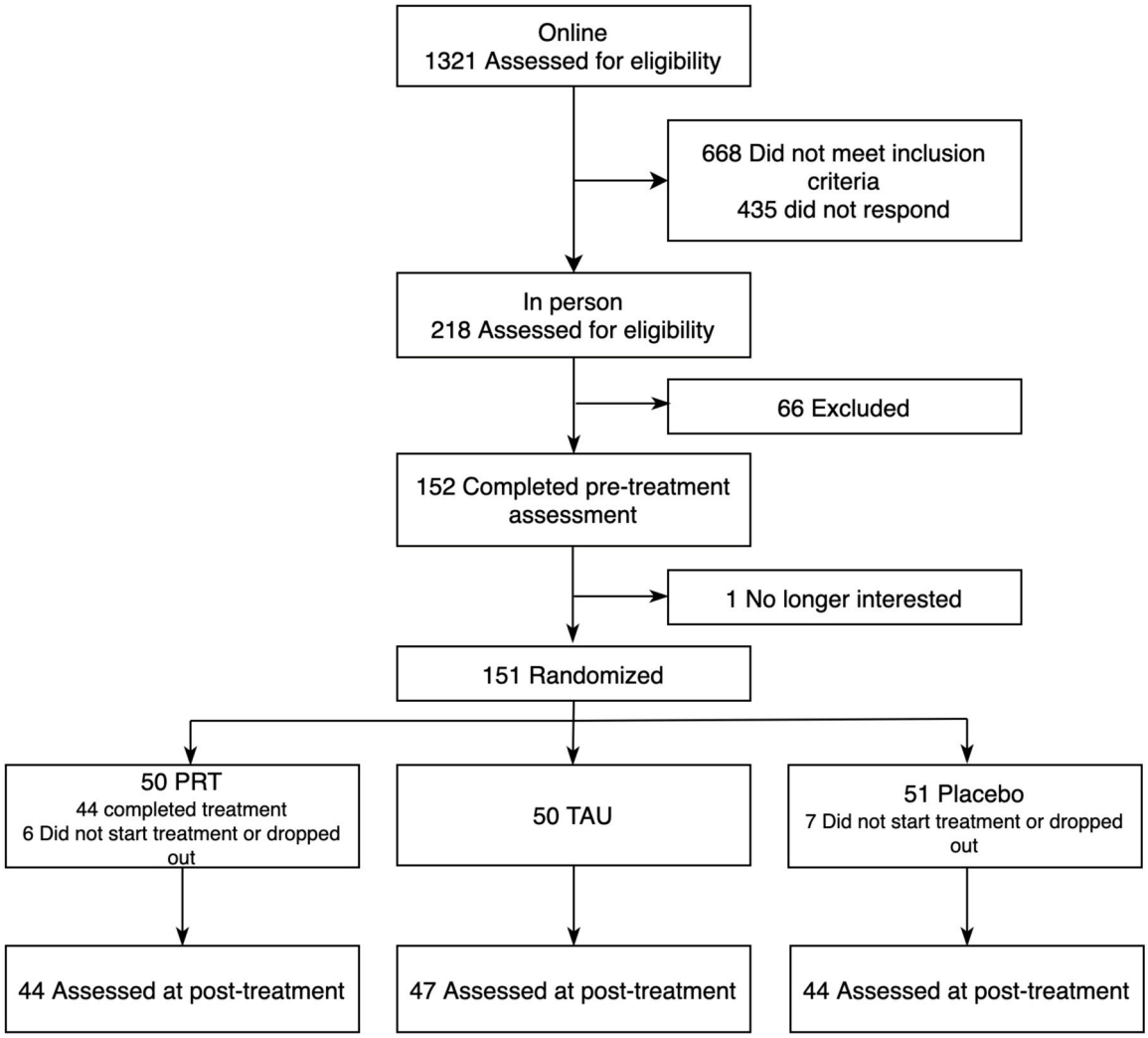


Fig. 2. CONSORT Participant Flow Diagram.

Table 1
Participants' descriptive data in the complete cases sample.

	TAU(n = 47)	Placebo(n = 44)	PRT(n = 44)	Total(n = 135)
Sex				
Male	22 (46.8%)	23 (52.3%)	19 (43.2%)	64 (47.4%)
Female	25 (53.2%)	21 (47.7%)	25 (56.8%)	71 (52.6%)
Age	41.7 (16.1)	39.6 (15.0)	45.1 (15.6)	41.9 (15.5)
Race				
Native American	1 (2.1%)	0 (0%)	0 (0%)	1 (0.7%)
Asian American	0 (0%)	2 (4.5%)	2 (4.5%)	4 (3.0%)
Black American	1 (2.1%)	2 (4.5%)	0 (0%)	3 (2.2%)
White American	40 (85.1%)	38 (86.4%)	42 (95.5%)	100 (74.1%)
Other or unknown	5 (10.6%)	2 (4.5%)	0 (0%)	7 (5.2%)
Pain duration (years)	10.2 (9.09)	8.27 (6.88)	11.4 (10.0)	10.8 (9.54)
Physical activity				
Almost none	3 (6.4%)	1 (2.3%)	6 (13.6%)	10 (7.4%)
1h/wk	8 (17.0%)	5 (11.4%)	3 (6.8%)	16 (11.9%)
3h/wk	14 (29.8%)	19 (43.2%)	14 (31.8%)	47 (34.8%)
7h/wk	20 (42.6%)	17 (38.6%)	19 (43.2%)	56 (41.5%)
≥14 h/wk	2 (4.3%)	2 (4.5%)	2 (4.5%)	6 (4.4%)
ODI (0−100) T0	21.9 (10.3)	19.0 (11.1)	24.2 (11.3)	22.5 (10.7)
ODI (0−100) T1	20.7 (10.7)	21.9 (10.9)	10.1 (10.6)	-
BPI (0−10) T0	3.51 (1.43)	3.89 (1.51)	4.02 (1.52)	3.79 (1.49)
BPI (0−10) T1	3.13 (1.45)	2.84 (1.64)	1.18 (1.24)	-

Note. Values show mean (standard deviation) or count (percentage). TAU, treatment as usual; PRT, Pain reprocessing therapy; ODI, Oswestry Low Back Pain Disability Questionnaire; BPI, Brief Pain Inventory Short Form; T0, pre-treatment; T1, post-treatment.

space and increase statistical power, GM analyses were conducted within a broad mask of regions most reliably associated with chronic pain-related GM adaptions in prior meta-analyses and regions linked to treatment response in prior studies.[5–7] The mask included the entire prefrontal cortex, the insula and several subcortical structures (i.e., thalamus, putamen, caudate, nucleus accumbens, pallidum, striatum, hippocampus, and amygdala).

Mediation analysis tested for (1) PRT vs. control differences in GMV changes (Treatment-mediator relationship, path a) controlling for pre-treatment confounders, (2) GMV changes associated with post-treatment clinical outcomes (Mediator-outcome relationship, path b) controlling for group allocation and pre-treatment confounders, and (3) GMV changes mediating the effects of PRT on post-treatment outcome compared to the control group (indirect effect, path ab). Four separate models tested effects on disability and pain intensity for PRT vs. placebo and PRT vs. TAU. While TFCE correction[73] provides more sensitivity to detect subtle differences in GM and is more robust to non-stationary data,[74,75] it is not yet implemented for mediation analysis. Thus, we applied an exploratory threshold of $p < 0.001$ uncorrected with a cluster extent $k > 50$ voxels for cortical regions and $k > 25$ for subcortical areas, similar to previous GM studies in chronic pain.[9,15] To improve robustness and reduce the risk of Type I errors, mediation effects were assessed using bootstrapping with 5000 resamples. Bootstrapped p-values were derived directly from the empirical null distribution, without relying on parametric T-statistics.

Due to inadequate statistical power, a treatment-mediator interaction term was not included in the outcome model of the mediation analysis.[76] Instead, we conducted post-hoc analyses to test for treatment-mediator interactions in clusters with a significant treatment-mediator and mediator-outcome relationships.[77] Additionally, we performed post-hoc analyses to test whether clusters observed in the treatment-mediator relationship for PRT vs TAU were also significantly different for PRT vs Placebo, and vice versa. Post-hoc analyses used summarized parameter estimates (beta coefficients) for GMV extracted for each participant using *marsbar*.[78]

3. Results

3.1. Participants

Of N=151 randomized, 135 completed post-treatment MRI, with drop-out due to participants not initiating treatment or withdrawing from treatment (Figure 2). Thus, the complete case sample consisted of 135 participants, of which 44 received PRT, 47 TAU and 44 Placebo (descriptive data in Table 1).

3.2. Brain mediation analysis

We found no evidence for mediation effect (path ab) for any of our models. Examining path-a and path-b separately, there were significant clusters associated with the treatment effects on the brain (treatment-mediator relationship, path a), and significant GMV changes associated with clinical outcomes (mediator-outcome relationship, path b), as described below. However, these results were non-overlapping, such that no clusters mediated the effect of treatment on clinical outcomes.

3.2.1. Disability as outcome

PRT vs treatment as usual: PRT led to large reductions in disability (ODI) relative to TAU (relative reduction of −12.64 points, 95% CI: −15.38, −9.91; $\eta^2 = -0.42$, 95% CI: −1.00, −0.29). Though no significant mediation was found, analyses revealed GMV increases in the anterior mid cingulate cortex (aMCC) for PRT vs. TAU (cluster size k = 73; x = −6, y = 30, z = 15) (Fig. 3**A**, Table 2). Analyses of the mediator-outcome relationship, testing how GMV changes relate to disability changes, identified five clusters. GMV reductions in the right and left thalamus (k = 63 and 67; x = 11, y = −32, z = 8; x = −11, y = −27, z = 11 respectively) and right amygdala (k = 37; x = 26, y = −2, z = −21) were associated with reductions in disability. GMV increases in the right dorsolateral prefrontal cortex (k = 59, x = −24, y = 41, z = 32) and the right ventrolateral prefrontal cortex (k = 281, x = 42, y = 47, z = −6) were associated with reduced disability (Table 2). Post-hoc analyses revealed a significant treatment-mediator interaction in right and left thalamus (Fig. 3**A**). In other words, the positive mediator-outcome relationship was mainly driven by the PRT group: GMV reductions after treatment were associated with reduced disability more strongly in patients allocated to PRT than TAU.

PRT vs open-label placebo: PRT also led to large reductions in disability relative to open label placebo (average relative reduction = −16.20; 95% CI: −19.16, −13.24; $\eta^2 = -0.46$, 95% CI: −1.00, −0.33). Patients who received PRT showed GMV increases in the left orbital part of the inferior frontal cortex (OFC; k = 63, x = −38, y = 21, z = −18) compared to placebo (Table 2). These changes, however, did not mediate PRT effects on disability (Fig. 3**B**). Three clusters were identified in the mediator-outcome relationship. Reduced disability was associated with decreased GMV in the right and left thalamus (k = 27 and 41; x = 8, y = −29, z = 8; x = −6, y = −21, z = 12 respectively) and increased GMV in the VLPFC (k = 99, x = 41, y = 47, z = −5). The post-hoc analysis revealed a significant treatment-mediator interaction in left thalamus (Fig. 3**B**). In other words, the positive mediator-outcome relationship was mainly driven by the PRT group: GMV reductions after treatment were associated with reduced disability more strongly in patients allocated to PRT than TAU.

3.2.2. Pain intensity as outcome

PRT vs treatment as usual: PRT led to large reductions in pain intensity relative to TAU (relative reduction = −2.23; 95% CI: −2.74, −1.74; $\eta^2 = -0.44$, 95% CI: −1.00, −0.31). As above, PRT vs. TAU led to GMV increases in the aMCC. These increases did not mediate PRT effects on pain intensity (Fig. 4**A**). Pain reductions were associated with one cluster: GMV reductions in the medial left thalamus (k = 127; x = −3, y = −20, z

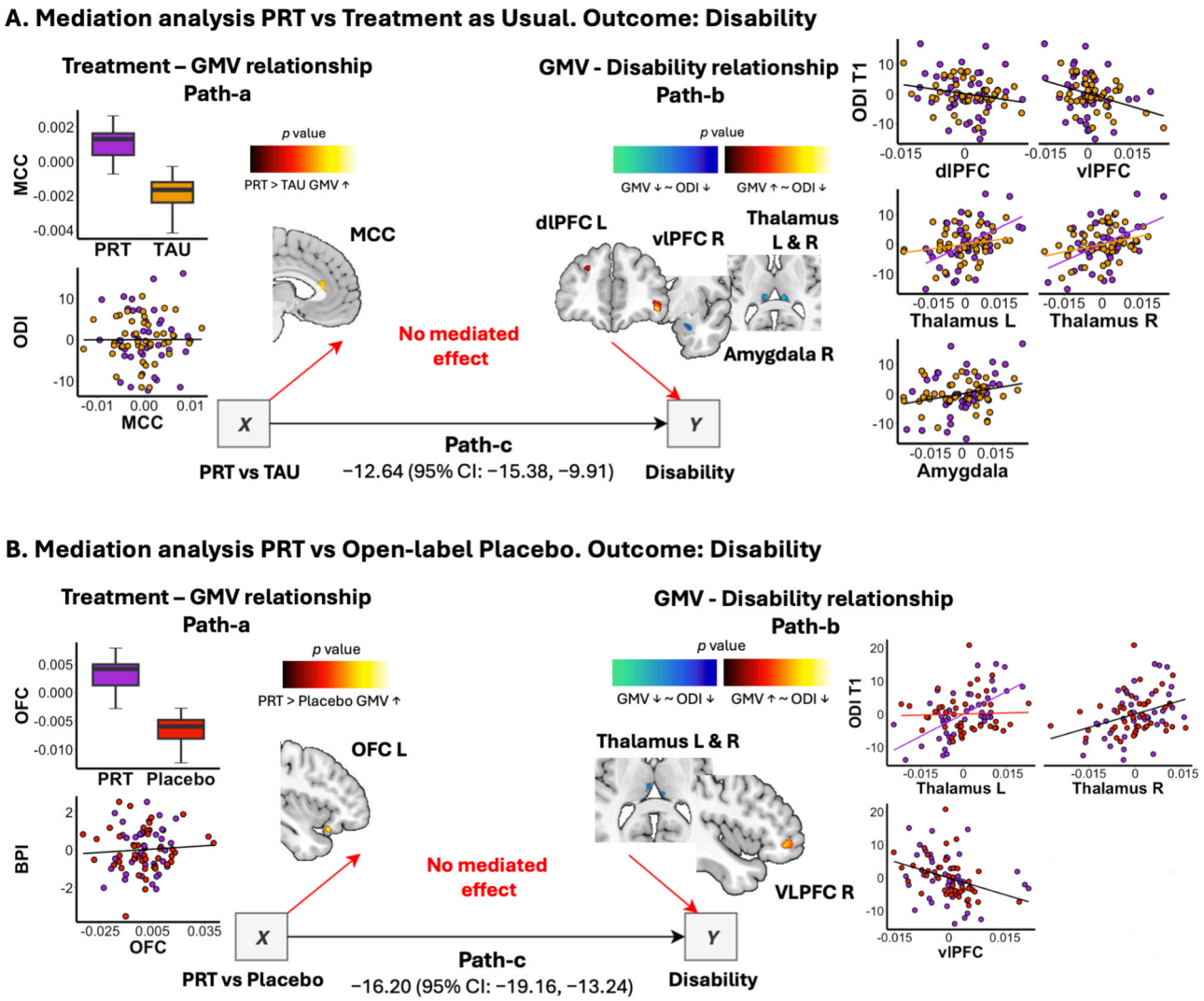


Fig. 3. (A) Mediation analysis for PRT vs TAU on disability (ODI). Results for the treatment-mediator effect are shown on the left and for the mediator-outcome effect on the right. There was no mediation effect. Boxplots depict changes in regional gray matter volume (GMV) by treatment condition. Scatterplots depict how changes in GMV are associated with clinical improvement (path *b*) by treatment condition, with model residuals displayed on the y axis. Post-hoc analyses found a significant treatment-mediator interaction in the left and right thalamus, indicating that the association between GMV reduction here and clinical improvement was driven mainly by PRT participants. The residuals are displayed in the y and x axis. **(B) Mediation analysis for PRT vs Open-Label Placebo on disability (ODI)**. There was no mediation effect. Boxplots depict changes in regional GMV by treatment condition. Post-hoc analyses found a significant treatment-mediator interaction in the left thalamus, indicating that the association between GMV reduction here and clinical improvement was driven mainly by participants in the PRT condition. PRT, pain reprocessing therapy; TAU, treatment as usual; GMV, gray matter volume; T0, pre-treatment; T1, post-treatment; MCC, midcingulate; dlPFC, dorsolateral prefrontal cortex; vlPFC, ventrolateral prefrontal cortex; OFC, orbitofrontal cortex.

= 9) (Table 2). This relationship did not differ between groups.

PRT vs open-label placebo: PRT led to large reductions in pain intensity relative to placebo (-1.76, 95% CI: −2.34, −1.19; η^2 = −0.30, 95% CI: −1.00, −0.17). As above, PRT vs. Placebo led to GMV increases in the OFC. These GMV increases did not mediate PRT effects on pain intensity compared to placebo. No clusters survived for the GMV – pain intensity relationship across PRT and placebo conditions (Fig. 4**B**).

3.3. Between-group differences in GMV: Post-hoc analyses for the treatment-mediator relationship

As each model included only two treatment conditions, we conducted post-hoc analyses to investigate effects across all three conditions in regions showing a PRT vs. placebo effect (the OFC) or PRT vs. TAU effect (the aMCC). Participants who received PRT also had significant GMV increases in the aMCC relative to placebo (0.003; 95% CI: 0.001, 0.006; p = 0.004) and marginally significant GMV increases in OFC relative to TAU (0.005; 95%CI: −0.000, 0.011; p = 0.059).

4. Discussion

People with chronic pain, including back pain, exhibit GM reorganization. Prior studies have shown that these GM alterations may change with treatment. However, prior trials either lacked control groups [35–37] or, when they included them, did not detect significant treatment-versus-control differences.[25,38,39] And, prior studies largely did not investigate how treatment-specific changes may relate to improvements in clinical outcomes. Thus, the scientific and clinical implications of observed longitudinal GM changes in prior studies remains unclear.

This study provides the first insights into how treatment may affect GM to improve chronic pain, by applying a VBM mediation analysis framework. Our findings indicate that, compared to TAU and open-label placebo, PRT increases GMV in key pain modulatory regions such as the

Table 2
Results mediation analyses. Cluster information.

	P value		Direction effect	Size	MNI coordinates			Anatomical location
	Min	Max			X	Y	Z	
Mediation analysis for disability (ODI). PRT vs TAU								
Treatment–Mediator	2.0734e-05	0.0009	PRT > TAU GM ↑	73	-6	30	15	aMCC L
Mediator-Outcome	4.8147e-07	0.0009	GM ↑ ~ ODI ↓	59	-24	41	32	dlPFC R
	4.1465e-22	0.0009	GM ↑ ~ ODI ↓	281	41	47	-6	vlPFC R
	3.414e-05	0.0009	GM ↓ ~ ODI ↓	63	11	-32	8	Thalamus Pulvinar R
	2.5975e-05	0.0009	GM ↓ ~ ODI ↓	67	-11	-27	11	Thalamus Pulvinar L
	0.0001	0.0009	GM ↓ ~ ODI ↓	37	26	-2	-21	Amygdala R
Mediation analysis for disability (ODI). PRT vs Placebo								
Treatment–Mediator	3.4286e-07	0.0009	PRT > Placebo GM ↑	63	-38	21	-18	OFC L
Mediator-Outcome	1.8405e-05	0.0009	GM ↑ ~ ODI ↓	99	41	48	-3	vlPFC
	0.00015	0.0009	GM ↓ ~ ODI ↓	27	8	-29	8	Thalamus Pulvinar R
	0.00013	0.0009	GM ↓ ~ ODI ↓	41	-6	-21	12	Thalamus Pulvinar L
Mediation analysis for pain intensity (BPI). PRT vs TAU								
Treatment–Mediator	3.159e-06	0.0009	PRT > TAU GM ↑	73	-6	30	15	aMCC
Mediator-Outcome	9.0719e-05	0.0009	GM ↓ ~ BPI ↓	127	-3	-20	9	Thalamus medial L
Mediation analysis for pain intensity (BPI). PRT vs Placebo								
Treatment–Mediator	3.0201e-07	0.0009	PRT > Placebo GM ↑	63	-38	21	-18	OFC L

Note. Significant clusters for treatment-mediator (effects of treatment on GMV) and mediator-outcome (associations between GMV changes and clinical improvement) relationships. Anatomical region labels are derived from the CanlabCore 2024 atlas. The minimum and maximum p-value for each cluster is provided; T values are not available as p-values were derived directly via bootstrapping of parameter estimates (see Methods). TAU, treatment as usual; PRT, Pain reprocessing therapy; ODI, Oswestry Low Back Pain Disability Questionnaire; BPI, Brief Pain Inventory Short Form; L, Left; R, Right; aMCC, anterior Midcingulate cortex; dlPFC, dorsolateral prefrontal cortex; vlPFC, ventrolateral prefrontal cortex; OFC, orbitofrontal cortex.

aMCC and the OFC. However, neither of these GM changes mediated the effects of treatment on disability or pain intensity. Instead, clinical improvement was associated with GM reductions in the thalamus and amygdala and GM increases in lateral prefrontal cortex. However, there were no significant PRT vs. control effects in these areas associated with clinical improvement. Thus, while treatment led to GMV changes, and GMV changes were associated with clinical improvement, our findings suggest that the treatment-specific effects on GMV do not explain clinical improvement following PRT.

We observed treatment-induced increases in GM in the aMCC and OFC. This aligns with prior meta-analytic findings of reliable GM decreases in the aMCC and the OFC for chronic pain samples relative to healthy controls.[5–7] The aMCC and OFC increases observed here following PRT thus suggest that treatment may remediate GMV deficits in chronic pain. Further, the aMCC is one of the core regions engaged in nociceptive processing, and may play a role in evaluating pain unpleasantness specifically, by integrating nociceptive input with cognitive evaluations through top-down mechanisms.[79–81] The OFC, on the other hand, is involved in the generation and regulation of emotional appraisals and, more specifically is related to the encoding of pain-related fear.[72,82] As PRT was superior to the control arms in reducing pain-related fear and threat appraisals (reported in the primary outcomes paper from the trial[49]), GM changes in these two regions could reflect, to some extent, an enhancement of cognitive and emotional control of pain after PRT. At the same time, as GMV changes in these regions were not associated with clinical improvement, these GMV changes may be an "incidental" effect of treatment with limited clinical relevance.

We found that GMV increases in the dlPFC and the vlPFC were associated with reductions in disability, regardless of the intervention received. The dlPFC and vlPFC are top-down modulatory regions where GM reductions have been consistently reported in people with CBP.[11,17,68,70,83,84] The dlPFC is actively involved in processing sustained pain and contributes to cognitive control and attention regulation of nociceptive stimuli.[85–87] The VLPFC, like the OFC, modulates emotions and is engaged by instructed reappraisal of emotional stimuli.[72,88] Several cohort studies have observed that with pain remission, GM within these two areas tends to normalize.[14–17] Yet, evidence from randomized trials including our findings here has not found treatment-specific effects in these regions.[25,38,39] As growing evidence suggests that the dlPFC and the VLPFC are involved in treatment expectations and placebo analgesia,[89–91] changes in these areas could potentially reflect shared mechanisms across interventions (e.g., contextual factors and factors associated with placebo response) and/or the natural history of recovery from chronic pain.[92]

Improvements in disability were also correlated with GMV decreases in the posterior thalamus (pulvinar nuclei) and amygdala and pain relief was linked to GMV decreases in the medial thalamus. GM adaptations to chronic pain in the amygdala and the thalamus have been extensively reported, though the evidence is conflicting about the direction of these effects (i.e., increase vs. decrease).[5,7] The amygdala plays a crucial role in the emotional evaluation of pain and in central sensitization.[93] The medial portion of the thalamus is involved in the processing of painful stimuli, processing and transmitting nociceptive input from the spinal cord to the cortex.[81,94] In chronic pain, abnormalities in the thalamus have been linked to the presence of central sensitization phenomenon,[95,96] including greater GMV in the medial portion of the thalamus associated with higher levels of pain in patient populations.[7,12] Our study extends these findings by showing that pain relief leads to decreased GMV of this thalamic subdivision, regardless the intervention received. On the other hand, the pulvinar thalamus and the amygdala are closely related components of the emotional-affective neural processing.[97] The pulvinar is involved in regulating emotional responses and has significant connections with brain regions like the amygdala and prefrontal cortex, which are crucial for processing emotions and stress. Interestingly, the relationship between GMV in the pulvinar thalamus and disability after treatment was only observed in patients who received PRT. While this pattern may align with theoretical models suggesting that PRT targets emotional processing and regulation, particularly within thalamo-cortical networks, we acknowledge that this interpretation is speculative and not directly supported by the mediation analyses. These findings should therefore be viewed as exploratory and hypothesis-generating.

The underlying physiological mechanisms responsible for the observed GMV changes in chronic pain are still not fully understood. Increasing evidence indicates that alterations in GMV may be influenced by neurogenesis, gliogenesis (i.e., changes in the number of non-neuronal cells), dendritic remodeling, vascular changes, and synaptic plasticity, among other factors. [98–103] These processes are thought to reflect local tissue and vascular adjustments driven by energy demands following neuronal activity, potentially contributing to long-term morphological changes. [101,104] However, it is important to note that

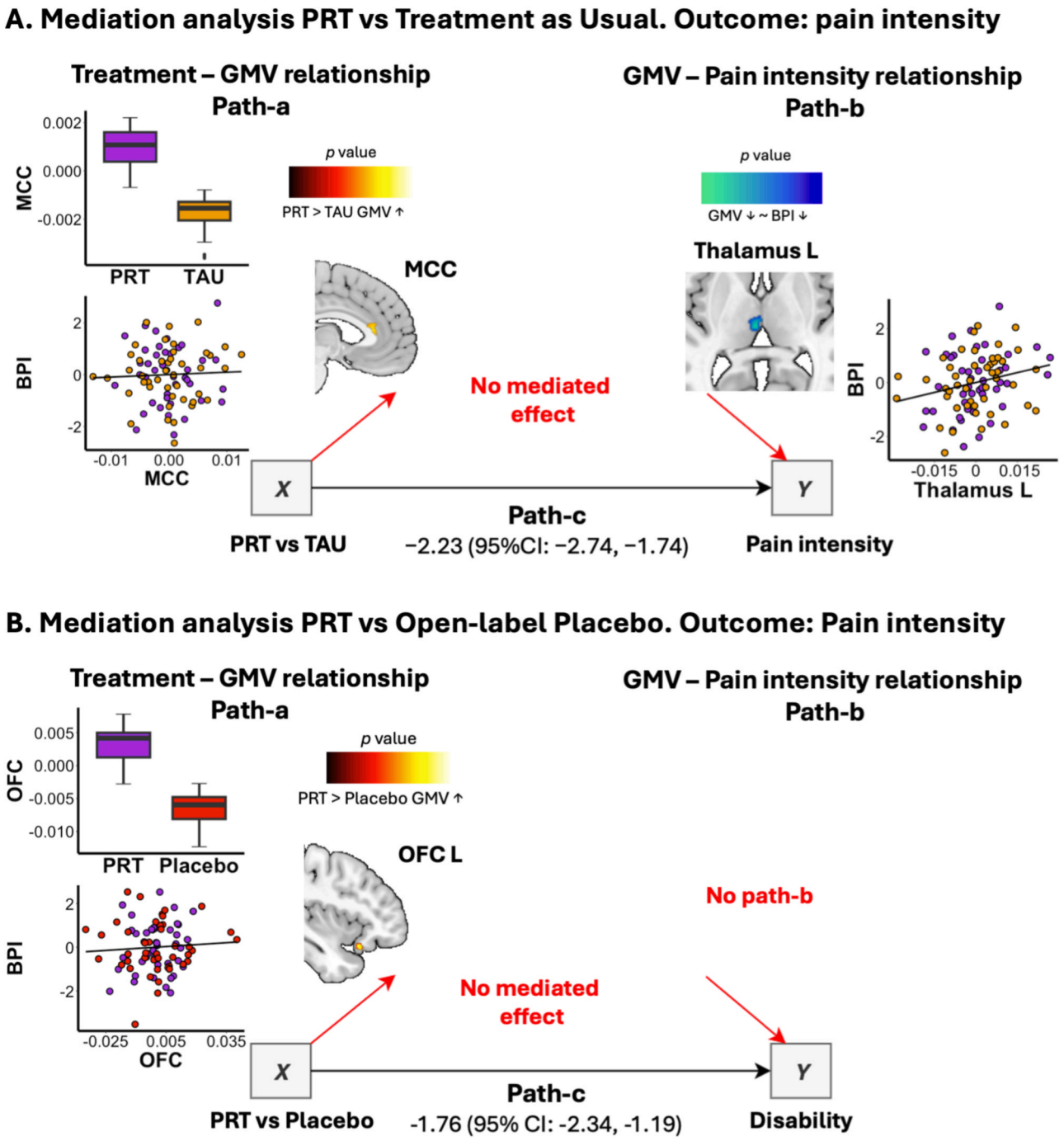


Fig. 4. (A) Mediation analysis for PRT vs TAU on pain intensity (BPI). There was no mediation effect with pain intensity as an outcome, similar to findings with disability as an outcome (Fig. 3). Results for the effect of treatment on GMV (treatment-mediator effect, path *a*) are shown on the left and for the assocation between GMV and clinical improvement (mediator-outcome effect, path *b*) on the right. Post-hoc analyses to test for treatment-mediator interactions in the left thalamus (scatter plot) found the assocation was similar in both treatment conditions. **(B)** Mediation analysis for PRT vs Open-label placebo on pain intensity. There was no mediation effect. Effects of treatment on GMV are shown on the left. No significant associations between changes in pain intensity and GMV were observed for PRT and placebo participants.

longitudinal changes in GMV as detected by MRI are inherently ambiguous and do not provide a direct measure of specific biological mechanisms. Rather than corresponding directly to specific cellular events, such changes likely reflect a complex and dynamic interplay of multiple neurobiological processes. Accordingly, in the context of psychological treatment, GMV alterations should be viewed as indirect markers of brain plasticity or functional adaptation, rather than conclusive evidence of any single underlying mechanism. Future research should also investigate how structural changes relate to changes in function (activity and connectivity).

Our findings do not provide evidence that changes in gray matter volume mediate the effects of PRT on clinical outcomes, suggesting that GMV alterations, at least in this sample, may not represent a primary mechanism of treatment response. However, this does not rule out the clinical relevance of GMV alterations in other contexts. Rather, it reflects the complexity of structural brain changes in chronic pain and the ongoing uncertainty about their role in recovery. While some studies suggest that GMV changes reverse following peripheral treatments like

surgery—implying a consequence of nociceptive input—others propose that centrally acting interventions may drive neuroplastic adaptations. Our findings do not resolve this debate but highlight the need for more mechanistically focused research.

At the same time, several limitations bear on the interpretation of our results. First, this secondary mediation analysis was not pre-specified, and it was likely underpowered and susceptible to some confounding bias (e.g., omission of important pre-treatment confounders). Factors such as depression, medication use, or expectancy effects, among others, could have confounded the relationship between treatment response and brain plasticity, and should be considered in future research. The sample was predominantly White, and future research should recruit more diverse participants and consider race/ethnicity as potential confounders. Second, our limited sample size also precluded the inclusion of treatment × mediator interaction term in the model, and we had to rely on exploratory post hoc analyses instead. We also used a relatively liberal statistical threshold ($p < 0.001$ uncorrected) which increases the risk of false positives and warrants cautious interpretation of the results. Third, it should be acknowledged that VBM has limitations in analyzing subcortical structures, including reduced sensitivity to subtle subregional changes, susceptibility to poor registration, and potential for false positives. Therefore, these results are best viewed as preliminary and hypothesis-generating rather than definitive evidence against mediation and future research should attempt to address these shortcomings.

5. Conclusions

Our findings indicate that, compared to treatment as usual and open-label placebo, PRT increases GMV in key pain modulatory regions such as the aMCC and the OFC shown to have reduced GMV in prior studies. However, these treatment-induced GM changes did not mediate reductions in disability or pain intensity. And, clinical improvement were connected to GM changes in lateral prefrontal and subcortical areas, aligned with prior findings implicating these areas in chronic pain. However, we did not observe treatment-specific effects in these areas.

Thus, on one hand, our findings align well with prior research and speak to the relevance of GMV changes in chronic pain in its treatment. On the other hand, our findings resemble two pieces of a puzzle that do not fit together, and question the relevance of GMV changes in the treatment of chronic pain. Future investigations are needed that, as attempted here, try to link treatment vs. control effects, GMV changes, and clinical improvement to better understand the significance of GMV alterations in chronic pain.

Author contributions

Yoni Ashar and Tor Wager contributed to the conception and design of the study. The data and statistical analyses were performed by Carlos Murillo and Yoni Ashar. Carlos Murillo and Yoni Ashar had full access to all the data in the study and take responsibility of the integrity of the data and the accuracy of the data analyses. Carlos Murillo made the first draft of the paper. Yoni Ashar and Tor Wager revised this paper critically for important intellectual content, gave final approval of this version of the paper and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Funding

This study was funded by National Institutes of Health grants R01DA035484 (Dr. Wager) and R01MH076136 (Dr. Wager); National Center for Advancing Translational Sciences grant TL1-TR-002386 (Dr. Ashar); the Radiological Society of North America (Dr. Flood); German Research Foundation grant GE2774/1-1 (Dr. Geuter); the Psychophysiologic Disorders Association; the Foundation for the Study of the Therapeutic Encounter; and community donations. Dr. Ashar reports consulting fees from the Pain Reprocessing Therapy Center and from Lin Health. Dr. Wager serves as a scientific advisor to Curable.

Conflict of Interest

No other conflicts are reported.

Acknowledgements

This study was funded by National Institutes of Health grants R01 DA035484 (Dr Wager), R01 MH076136 (Dr Wager), National Center for Advancing Translational Sciences grant TL1-TR-002386 (Dr Ashar), Radiological Society of North America (Dr Flood), German Research Foundation grant GE 2774/1–1 (Dr Geuter), the Psychophysiologic Disorders Association, the Foundation for the Study of the Therapeutic Encounter, and community donations. The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication. Dr Ashar reports grants from the National Institutes of Health during the conduct of the study and personal fees from UnitedHealth Group, Lin Health, Inc, Pain Reprocessing Therapy Center, Inc, and Mental Health Partners of Boulder County outside the submitted work Dr Wager reports grants from the National Institutes of Health and the Foundation for the Study of the Therapeutic Encounter, and funding to support trainees from the Radiological Society of North America and the German Research Foundation; he is on the Scientific Advisory Board of Curable Health. No other disclosures were reported.

Data Availability

Data available upon reasonable request.

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